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Speculator activity and the cross-asset predictability of FX returns☆

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ABSTRACT

This paper tests the gradual information diffusion hypothesis, which suggests that information spreads gradually across asset markets, to explain the role of speculator activity in the cross-asset return predictability of foreign exchange (FX) market strategies. We argue that the activity of speculators increase the rate of information diffusion across asset markets. Hence, we expect the predictive effect from the equity and commodity markets on FX market strategies to be weaker when speculators are active in the FX market. Our results show that, when speculator activity is high, the equity market's ability to predict the FX market dissipates, but not to the same extent as for the commodity market. Our findings suggest that speculators play a vital role in enhancing informational efficiency in the FX market.

1. Introduction

The literature documents ample evidence that asset returns encompass a predictable component. Conventional finance theories, which assume investors to be unboundedly rational and to have unlimited information processing capacity, have, however, been ineffectual at explaining this predictability. Over the past decade, behavioural economics models have gained prominence in explaining the predictability of asset returns. In this paper, we use one such behavioural model, the gradual information diffusion hypothesis (Hong & Stein, 1999), to explain the role of speculators in the cross-asset return predictability of foreign exchange (FX) market strategies. Specifically, we examine the ability of the equity and commodity markets to predict the FX market with respect to speculative activity in the FX market.

The gradual information diffusion hypothesis posits that economic information flows slowly across asset markets and generates return predictability. This is a consequence of investors' inability to process every piece of information. The literature shows that the impact of economic fundamentals is pervasive across asset markets.[1] In addition, evidence on predictability within and across asset classes has been shown to be consistent with the gradual information diffusion hypothesis (e.g. Chordia & Swaminathan, 2000; Cohen & Frazzini, 2008; Driesprong, Jacobsen, & Maat, 2008; Hong, Lim, & Stein, 2000; Hong, Torous, & Valkanov, 2007; Hou, 2007; Lu & Jacobsen, 2016; Menzly & Ozbas, 2010; Rapach, Strauss, & Zhou, 2013). This research has, however, not given due consideration to the role of speculators as catalysts in the dissemination of information across economically linked asset markets. Understanding the role of speculators is imperative, because their activities can enhance information diffusion between markets as the participants actively incorporate the relevant information. This paper, therefore, explores the role of speculators as a fundamental driver in the gradual information diffusion process.

To the extent that speculators are informed investors,[2] they should spread information more rapidly than uninformed investors (Menzly & Ozbas, 2010). Thus, investigating the cross-asset predictability of FX market strategies in conjunction with the activity of FX market participants provides insights into the important question of the role of

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[1] See, for example, Chen and Rogoff (2003), Kearns (2007), Campbell, Serfaty-De Medeiros, and Viceira (2010), and Lustig et al. (2014).

[2] We acknowledge that speculators are not necessarily informed. However, the speculator data used in this study is characterized by large traders, such as hedge funds, commodity trading advisors, and money managers, who are generally assumed to be informed. For example, Menzly and Ozbas (2010) present evidence that institutional investors trade like informed investors.

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market speculators in the information diffusion process. In addition, by distinguishing the role of speculators, this paper integrates several strands of the financial literature: FX market-related return predictability (e.g. Bakshi & Panayotov, 2013; Lu & Jacobsen, 2016), the role of speculators in the FX market (e.g. Fong, 2013), and the role of speculators in the information diffusion process (Hong & Stein, 1999; Menzly & Ozbas, 2010).

The behaviour of FX market participants is state dependent. For example, Fong (2013) documents a significant increase in hedge funds' carry trade positions in *hot markets* (market states with very high currency returns) and notes that optimism in the stock market has a spillover effect on hedge fund speculation in the FX market. Additionally, evidence in the literature suggests that, as informed traders, hedge funds' activity increases market efficiency (Kokkonen & Suominen, 2015; Menzly & Ozbas, 2010). Assuming that speculators, as informed investors, monitor market information and process information exceptionally well, a larger number of speculators, or a greater extent of their activity, should increase the rate of information diffusion into the market. Therefore, we hypothesize that predictive effect from the equity and commodity market returns on FX strategy returns is weaker when FX speculators are active. This hypothesis implies that the gradual information diffusion between the equity and commodity markets and the FX market is dependent on speculative activity in the FX market. The FX strategies we consider are the carry trade, momentum, and fundamental strategies, which are well known both in the literature and in practice (e.g. Kroencke, Schindler, & Schrimpf, 2013).

We report several notable findings. First, we find that the equity market – Standard & Poor's (S&P) 500 – return predicts each strategy, and the commodity index predicts the majority of the short and long legs of all three FX strategies considered in this paper. In relation to speculator activity, we find that, in line with our hypothesis, the predictive ability of equity market returns is nonsignificant when speculators are active in the Australian dollar (AUD)/Japanese yen (JPY) carry trade. However, speculator activity does not appear to alter the commodity market's predictive ability. Additionally, the predictability of FX strategy returns peaks after five to 10 weeks and decreases thereafter, but only in periods characterized by low speculator activity. These findings lend credence to our hypothesis and suggest that speculator activity increases the rate of information diffusion from the equity market to the FX market.

Our key contribution is that we recognize FX market participant activity as an important and novel aspect of asset return predictability. To the best of our knowledge, the literature has not yet addressed the role of speculators in the information diffusion process, thus leaving a gap in the literature.

The remainder of our paper is organized as follows. Section 2 presents the data, the FX strategies, and the variables. Section 3 explains the methodology and discusses the results. Section 4 provides further analysis. Section 5 concludes the study.

2. Data and methodology

2.1. Sample construction

We are guided by the literature in our choice of cross-asset predictors in our empirical analysis. We include the two broad indexes in the equity and commodities markets, the S&P 500 Equity Index and the Commodity Research Bureau (CRB) Spot Index, respectively, as the main research variables in predicting FX strategy returns (e.g. Bakshi & Panayotov, 2013; Chen, Rogoff, & Rossi, 2010; Groen & Pesenti, 2009; Kearns, 2007; Lu & Jacobsen, 2016). We also include currency volatility[3] and liquidity[4] as additional control variables in our empirical analyses to control for predictability that could be related to risk premia (e.g. Brunnermeier, Nagel, & Pedersen, 2009; Menkhoff, Sarno, Schmeling, & Schrimpf, 2012). Following the literature, we normalize every predictor used over a three-month period, computed as the percentage change in the variables from month t - 4 to month t - 1, divided by three, before including them in the regressions (e.g. Bakshi & Panayotov, 2013; Fama & French, 1988).[5]

The data on the price series for the S&P 500 index and the CRB Spot Index, as well as non-seasonally adjusted consumer price indices (CPI) with the base year 2010, are obtained from Datastream. We also obtain the daily bid, mid, and ask spot rates and one-month forward rates for the G-10 currencies – the AUD, Canadian dollar (CAD), euro (EUR), JPY, the Norwegian krone (NOK), the New Zealand dollar (NZD), the Swedish krona (SEK), the Swiss franc (CHF), the British pound (GBP), and the German mark (DM) per U.S. dollar (USD) – from the same source. Following common praxis, we use the DM until the introduction of the EUR in January 1999. Analogously, the German CPI is used for the currency and forward data before January 1999; thereafter, we switch to the euro area CPI. Finally, we access the TED spread via the Federal Reserve Bank of St. Louis webpage.[6] We use monthly data and our full data sample period spans from January 1997 to December 2015.[7]

We collect data on trader positions in futures contracts in the currency market from the U.S. Commodity Futures Trading Commission's (CFTC) Commitment of Traders report obtained via the CFTC website.[8] These data have typically been used to measure speculative capital in the currency market and to proxy for carry trade positions (e.g. Brunnermeier et al., 2009; Fong, 2013). The positions included in the reports correspond to 70–90% of a market's total open interest.[9] The CFTC reports the trading positions of two trader groups, *commercial* and *noncommercial* traders, where the former (latter) refer to traders using the market for hedging (speculative) purposes. We focus on the latter group and refer to them as speculators. The data from the CFTC do not cover all of the G-10 currencies, but the CFTC data enable us to obtain reliable series for AUD, GBP, CAD, EUR, JPY and CHF against the USD. We use end-of-month observations to construct our activity measures.

2.2. Variables

2.2.1. Return calculation

We follow Menkhoff et al. (2012) in calculating currency returns. Let s_t and f_t be the logarithm of the spot rate of the foreign currency per USD and the logarithm of the one-month-forward contract on the same currency, respectively. The returns from buying a foreign currency, C, in the forward market and selling it in the spot market are then

$$r_{t+1}^{C} = f_t^{C} - s_{t+1}^{C}. \tag{1}$$

Eq. (1) can be rewritten in terms of the forward discount/premium, as follows:

$$r_{t+1}^{C} = f_t^{C} - s_t^{C} - \Delta s_{t+1}^{C}. \tag{2}$$

The forward discount/premium approach implies that covered

[3] Following Bakshi and Panayotov (2013), we construct monthly currency volatility as the square root of the sum of squares of daily log changes in the exchange rates, averaged over the set of currencies.

[4] Following Brunnermeier et al. (2009) and Byrne, Ibrahim, & Sakemoto et al. (2018) we use the TED Spread to proxy for liquidity.

[5] For robustness, we also run the entire analysis without normalization, that is, with simple one-month lagged variables as predictors. The results are qualitatively the same and lend even stronger support for our hypothesis than in the normalized case. We also normalize only the two predictors of particular interest, namely, the S&P 500 and CRB indices, and find almost exactly the same results as in our main analysis.

[6] https://fred.stlouisfed.org/.

[7] This period excludes some variables that use lagged values starting from January 1998 or September 1998 (see Table 1 for details).

[8] See https://www.cftc.gov.

[9] See http://www.cftc.gov/MarketReports/CommitmentsofTraders/ExplanatoryNotes/index.htm.

interest rate parity holds, taking into account interest payments. Covered interest rate parity states that the forward discount is approximately equal to the interest rate differential:

$$f_t^C - s_t^C \approx i_t^C - i_t, \tag{3}$$

where i_t^C is the foreign interest rates and i_t is the U.S. interest rate. Equivalently, currency returns can be written in terms of the interest rate differential as

$$r_{t+1}^C \approx i_t^C - i_t - \Delta s_{t+1}^C. \tag{4}$$

Thus, currency returns capture changes in the spot rates, as well as the interest rate component.

We consider transaction costs, as Menkhoff et al. (2012), by calculating the returns to a long position and a short position with the following equations, respectively:

$$r_{t+1}^{C,Long} = f_t^{C,b} - s_{t+1}^{C,a} \tag{5}$$

and

$$r_{t+1}^{C,Long} = -f_t^{C,a} + s_{t+1}^{C,b}. \tag{6}$$

where the superscripts a and b indicate ask and bid prices, respectively. As in Lustig, Roussanov, and Verdelhan (2014), the return to a long position represents an investor who buys the foreign currency (or sells the USD forward contract at the bid price) at time t and sells the foreign currency (buys the USD spot at the ask price) in the spot market at time $t + 1$, and vice versa for the return to a short position.

If a position is held for more than one continuous month, forward positions would have to be re-entered every month (since they have a maturity of one month), thus incurring transaction costs. A spot rate trade can, however, be postponed until an exit in the position. The return is thus calculated as below.

$$r_{t+1}^{C,Long} = f_t^{C,b} - s_{t+1}^C \; and \; r_{t+1}^{C,Long} = -f_t^{C,a} + s_{t+1}^C \tag{7}$$

Lyons (2001) reports that the bid–ask spreads from Reuters are approximately double the values of interdealer spreads. Our return calculations are therefore conservative.

2.2.2. Carry trade strategy

The carry trade strategy is executed by borrowing low-yield currencies and buying high-yield currencies. We construct a carry trade strategy from the perspective of a U.S. investor who takes long positions in currencies with a positive forward premium, that is, a positive interest rate differential, and short positions in currencies with a forward discount, that is, a negative interest rate differential vis-à-vis the USD. The forward premium is defined as

$$F_t^{C,Premium} = f_t^C - s_t^C. \tag{8}$$

The decision variable, d_t^C, of going long (+1), short (−1), or not investing (0) in foreign currency C is written as follows:

$$d_t^C = \begin{cases} +1 \text{ if } F_{t-1}^{C,Premium} > 0 \\ -1 \text{ if } F_{t-1}^{C,Premium} < 0 \\ 0 \text{ if } F_{t-1}^{C,Premium} = 0 \end{cases} \tag{9}$$

We distinguish between the short and long legs of the strategy, and the returns for the corresponding legs are calculated using the following equations, respectively:

$$r_t^{Carry,Long} = \frac{1}{N}\sum_{C=1}^{N} d_t^C \cdot r_t^{C,Long} \tag{10}$$

and

$$r_t^{Carry,Short} = \frac{1}{N}\sum_{C=1}^{N} d_t^C \cdot r_t^{C,Short}, \tag{11}$$

where N is the number of currencies in the portfolio and r_t^C is the currency return.

2.2.3. Fundamental strategy

The fundamental strategy takes positions in currencies based on their value in relation to some fundamental benchmark. An interesting feature of the fundamental strategy is that it yields high returns when volatility is high, unlike the carry trade strategy (Copeland & Lu, 2016). Similarly to Copeland and Lu (2016), we evaluate the currencies in relation to their long-run average real exchange rate. The real exchange rate of a currency is given by

$$q_t^C = s_t^C + (p_t - p_t^C), \tag{12}$$

where s_t^C is the logarithm of the exchange rate per USD of currency C, p_t^C is the CPI of the foreign country, and p_t is the U.S. CPI. We calculate the real exchange rate based on mid-quotes. We follow Jordà and Taylor (2012) and define the long-run equilibrium real exchange rate $\bar{q}_t^C$ as the three-year rolling average of the real exchange rate.[10] A long (short) position is established if the real exchange rate is lower (higher) than the long-run equilibrium real exchange rate. Formally, the decision variable z_t^C is constructed as

$$z_t^C = \begin{cases} +1 \; if \; q_{t-1} < \bar{q}^C{}_{t-1} \\ -1 \; if \; q_{t-1} > \bar{q}^C{}_{t-1} \\ 0 \; if \; q_{t-1} = \bar{q}^C{}_{t-1} \end{cases} \tag{13}$$

We distinguish between the short and long legs of the strategy and calculate the respective returns for the corresponding legs as follows:

$$r_t^{Fund,Long} = \frac{1}{N}\sum_{C=1}^{N} z_t^C \cdot r_t^{C,Long} \tag{14}$$

and

$$r_t^{Fund,Short} = \frac{1}{N}\sum_{C=1}^{N} z_t^C \cdot r_t^{C,Short}, \tag{15}$$

where N is the number of currencies in the portfolio and r_t^C is the return of currency C, as explained above.

2.2.4. Momentum strategy

The momentum strategy buys past winners and sells past losers. We include the momentum strategy in our analyses, since institutional traders have previously been found to be momentum traders in the FX market (e.g. Fong, 2013). Consistent with the work of Menkhoff et al. (2012), the momentum strategy that we are considering uses a 12-month formation period, a one-month skip period, and a one-month holding period. The long/short decision, x_t, is formalized as

$$x_t^C = \begin{cases} +1 \; if \; r_{t-12:t-1}^{C,Cumulative} > 0 \\ -1 \; if \; r_{t-12:t-1}^{C,Cumulative} < 0 \\ 0 \; if \; r_{t-12:t-1}^{C,Cumulative} = 0 \end{cases} \tag{16}$$

where $r_{t-12:t-1}^{C,Cumulative}$ is the cumulative return from month t - 12 to month t - 1 for currency C. We distinguish the short strategy from the long leg of the strategy and estimate the returns for the corresponding leg as follows:

$$r_t^{Mom,Long} = \frac{1}{N}\sum_{C=1}^{N} x_t^C \cdot r_t^{C,Long}, \tag{17}$$

and

$$r_t^{Mom,Short} = \frac{1}{N}\sum_{C=1}^{N} x_t^C \cdot r_t^{C,Short}, \tag{18}$$

where N is the number of currencies in the portfolio and r_t^C is the return of currency C, as explained above.

2.2.5. Speculator activity

To measure speculation in the FX market we construct net open

[10] Note that Jordà and Taylor (2012) use a five-year window.

positions following Brunnermeier et al. (2009) as

$$NOI_t^C = \frac{Long_t^C - Short_t^C}{Total\ Open\ Interest_t^C}, \tag{19}$$

where C is the currency of interest. Thus, net open positions (NOI_t^C) are calculated as the long positions ($Long_t^C$) minus the short positions ($Short_t^C$) over the sum of the long and short positions ($Total\ Open\ Interest_t^C = Long_t^C + Short_t^C$), in line with Brunnermeier et al. (2009) and similar to Fong (2013). Since carry traders are typically long in the AUD and short in the JPY, we calculate the final measure of investor activity in the carry trade by subtracting the net open positions in the JPY from those in the AUD as follows:

$$POS_t^{audjpy} = (NOI_t^{aud} - NOI_t^{jpy})/2. \tag{20}$$

In Eq. (20), higher values of POS_t^{audjpy} indicate increased carry trade activity.[11] We focus on activity in the AUD and JPY, two major carry trade currencies, but include the other relevant currencies in robustness tests.[12] We use a dummy variable for activity that takes the value of one when the standardized activity measure is positive, and zero otherwise. Thus, the dummy variable indicates high- and low-activity periods reflecting periods of inflow and outflow in a similar spirit as Plantin and Shin (2018) and Stambaugh, Yu, and Yuan (2012).

2.3. Descriptive statistics

Panel A of Table 1 presents the descriptive statistics of the S&P 500 index returns (Δ*SP500*), the CRB Spot Index returns (Δ*CRB*), and the long and short FX strategy returns. The descriptive statistics show that the annualized mean returns are positive for every strategy and their legs, with the exception of the fundamental strategy's long leg. The annualized mean returns of the long leg of the carry trade (*CTLong*) and the short leg of the carry trade (*CTShort*) are 1.19% and 1.03%, respectively. These results are comparable to the returns presented by Bakshi and Panayotov (2013), Filipe and Suominen (2015), and Lu and Jacobsen (2016), where the annualized returns of a combined high-minus-low strategy ranges between 2% and 4%. The Sharpe ratios of 0.10 and 0.11 for the long and short legs of the carry trade, respectively, are on the lower side but within the range presented for combined high-minus-low carry trade strategies by Bakshi and Panayotov (2013), Lustig et al. (2014), and Filipe and Suominen (2015). When previous studies find higher Sharpe ratios, they also use larger sets of currencies. The returns on the long and short legs of the fundamental strategy reported in this paper (−0.79% and 1.05%, respectively) are lower than those documented by Copeland and Lu (2016). This difference could be due to the different valuation method and the fact that the authors use a larger cross section of currencies.[13] We follow Jordà and Taylor (2012) and define the fundamental value as the three-year average of the real exchange rate (note that Jordà and Taylor (2012) use a five-year average). The annualized mean returns of the long and short legs of the momentum strategy (0.67% and 0.63%, respectively) are lower than roughly 6% of the combined momentum strategy presented by Menkhoff et al. (2012). We attribute this difference to the use of 48 currencies, including those of emerging markets, and a longer data period that includes the 1970s and 1980s.

The highest (lowest) annualized standard deviation of 15.69% (8.13%) is reported for the S&P 500 index (*FundLong*). The highest Sharpe ratio is achieved by the S&P 500 index (0.27). The returns of the long and short legs of the carry trade are negatively skewed, in line with previous research (e.g. Brunnermeier et al., 2009), while the fundamental strategy returns (earning money on reversals to fundamentals) are positively skewed. The descriptive statistics for the trader groups' activity variables are presented in Panel B of Table 1. The measure of speculator activity (POS_t^{audjpy}) in the AUD/JPY carry trade is standardized, and therefore has a mean of zero and a standard deviation of one.

Table 2 displays the monthly inclusion frequency for each currency in the different FX strategies. The descriptive statistics for *CTLong* show that the most frequent currencies for the long leg of the carry trade are the NZD and the AUD, which are included 78.02% and 65.09% of the months, respectively. Regarding the short leg of the carry trade strategy, the statistics for *CTShort* show that the two most frequent currencies are the JPY and the CHF, included 69.40% and 59.05% of the months, respectively. These four currencies are typical carry trade currencies. For the fundamental strategy, some currencies are overrepresented in both the long and short legs. The JPY, NOK, and SEK are included in the long leg in 56.5%, 34%, and 46.1% of the sample period, respectively, and in the short leg in 42%, 24.6%, and 34.9% of the sample period. For the momentum strategy, the set of included currencies is relatively diverse, but typical long carry trade currencies (AUD and NZD) are overrepresented in the long leg and typical short carry trade currencies (JPY and CHF) are overrepresented in the short leg.

Table 3 presents the correlations for the equity and commodity market returns, the FX strategy returns and FX volatility (Δ*FXVol*), and the funding liquidity (Δ*LIQ*) variables. All the long legs of the FX strategies are positively correlated with each other, as are all the short legs. Conversely, all the long legs of the FX strategies are negatively correlated with other strategies' short legs. All the long leg returns are also negatively correlated with Δ*FXVol* and Δ*LIQ* (although non-significantly for the long leg of the fundamental strategy), and the short legs of the fundamental and momentum strategy have a positive correlation with Δ*FXVol* and Δ*LIQ*. Equity returns are positively correlated with long leg carry trade returns (0.23) and commodity returns (0.39) and negatively correlated with both Δ*FXVol* (−0.4) and Δ*LIQ* (−0.36). Commodity returns are positively correlated with carry trade long leg returns (0.34) and momentum long leg returns (0.24) and negatively correlated with momentum short leg returns (−0.27) and short leg returns of the fundamental strategy (−0.34). Commodity returns are also negatively correlated with Δ*FXVol* (−0.32) and Δ*LIQ* (−0.35). Additionally, Δ*FXVol* and Δ*LIQ* are positively correlated, with a correlation coefficient of 0.28.

We evaluate the three-year rolling correlation between the two predictors S&P 500 index returns and CRB Spot Index returns in untabulated results. We observe a correlation close to zero until the financial crisis of 2008, when the correlation increases drastically to around 0.8, to only gradually decrease until the end of the sample period. The full-sample correlation coefficient is 0.39 (see Table 3). To avoid any multicollinearity problems, we follow, for example, Lu and Jacobsen (2016) and focus on the two predictors separately in the analysis.

3. Empirical analysis

3.1. Predictability of FX returns

To evaluate the predictability in the FX strategies, we estimate regressions of the following form:

$$y_t = \delta_0 + \mathbf{x}_{\mathbf{t-1}}'\delta_\chi + \epsilon_t, \tag{21}$$

where $y_t = r_t^{\text{Strategy, Leg}}$ is the return of an FX strategy, $\boldsymbol{\delta} = [\delta_0, \boldsymbol{\delta_\chi}']'$ is a vector of regression coefficents, and $\mathbf{x_{t-1}} = [\Delta SP500_{t-1}, \Delta CRB_{t-1}, \Delta FXVol_{t-1}, \Delta LIQ_{t-1}]$ represents the predictors and controls. Eq. (20) relates to testing the gradual information diffusion hypothesis without considering speculator activity to mirror the analysis of Lu and Jacobsen (2016). We start with univariate versions of Eq. (21) with

[11] Note that we standardize the final measure.

[12] In Section 4.1, we form a broader measure of carry trade using several currencies. The results remain similar to those in our main analysis.

[13] Copeland and Lu (2016) use a sample of 29 Organisation for Economic Cooperation and Development countries.

Table 1
Descriptive statistics This table presents the descriptive statistics for the main variables. Panel A shows the annualized returns, standard deviation, annualized Sharpe ratio (SR), skewness, the start and end dates for the variables, and the number of observations for the short and long legs of the FX strategies. The variables *CTShort* and *CTLong* are the long and short legs, respectively, of a carry trade strategy that takes long positions in currencies with high interest rates and short positions in currencies with low interest rates. The variables *FundLong* and *FundShort* are the returns of the long and short legs, respectively, of a fundamental strategy that takes long positions in undervalued currencies and short positions in overvalued currencies in relation to three-year average real exchange rates. The variables *MomShort* and *MomLong* are the returns of the short and long legs, respectively, of a momentum strategy with a 12-month formation period and a one-month skip period. The variable $\Delta SP500$ is the S&P 500 index return and ΔCRB is the CRB Spot Index return. Panel B presents the descriptive statistics for the variable of speculator activity in the AUD/JPY carry trade.

Panel A: strategy and predictor returns

Variable	Ann. Return	Std Dev	Sharpe Ratio	Skewness	Start	End	N
CTLong	1.19	11.36	0.10	−0.45	Jan 1997	Dec 2015	228
CTShort	1.03	9.22	0.11	−0.25	Jan 1997	Dec 2015	228
FundLong	−0.79	8.13	−0.10	0.42	Jan 1997	Dec 2015	228
FundShort	1.05	9.62	0.11	0.09	Jan 1997	Dec 2015	228
MomLong	0.67	9.76	0.07	−0.27	Jan 1998	Dec 2015	216
MomShort	0.63	10.07	0.06	0.24	Jan 1998	Dec 2015	216
$\Delta SP500$	4.17	15.69	0.27	−0.79	Jan 1997	Dec 2015	228
ΔCRB	0.77	10.89	0.07	−1.68	Jan 1997	Dec 2015	228

Panel B: Activity Measure

Variable	Mean	Std Dev	Median	Min	Max	Start	End	N	$POS_t^{audjpy} > 0$	$POS_t^{audjpy} \leq 0$
POS_t^{audjpy}	0	1	−0.05	−2.64	2.18	Jan 1997	Dec 2015	228	104	124

Table 2
Composition of FX Strategies (%) This table presents the inclusion frequency in percent (%) of each currency in the long and short legs of the carry trade strategy and fundamental and momentum strategies. The variables *CTShort* and *CTLong* are the long and short legs, respectively, of a carry trade strategy that takes long positions in currencies with high interest rates and short positions in currencies with low interest rates. The variables *FundLong* and *FundShort* are the returns of the long and short legs, respectively, of a fundamental strategy that takes long positions in undervalued currencies and short positions in overvalued currencies in relation to three-year average real exchange rates. The variables *MomShort* and *MomLong* are the returns of the short and long legs, respectively, of a momentum strategy with a 12-month formation period and a one-month skip period.

	AUD	CAD	EUR	JPY	NOK	NZD	SEK	CHF	GBP	DM
CTLong	65.09	5.60	3.02	0.00	20.26	78.02	8.62	0.43	18.97	0.00
CTShort	0.43	21.55	11.21	69.40	15.52	6.90	10.34	59.05	5.60	0.00
FundLong	0.00	9.48	15.09	56.47	34.05	2.59	46.12	9.05	27.16	0.00
FundShort	5.60	22.41	3.88	41.81	24.57	17.24	34.91	14.22	34.91	0.43
MomLong	32.27	19.55	5.00	22.73	20.00	37.73	11.36	21.82	25.45	4.09
MomShort	17.27	18.64	18.64	49.55	15.00	19.09	20.91	20.45	17.27	3.18

Table 3
Correlations This table presents the correlation statistics of the variables used in this study. The variables *CTShort* and *CTLong* are the long and short legs, respectively, of a carry trade strategy that takes long positions in currencies with high interest rates and short currencies with low interest rates. The variables *FundLong* and *FundShort* are the returns of the long and short legs, respectively, of a fundamental strategy that takes long positions in undervalued currencies and short positions in overvalued currencies in relation to three-year average real exchange rates. The variables *MomShort* and *MomLong* are the returns of the short and long legs, respectively, of a momentum strategy with a 12-month formation period and a one-month skip period. The variable $\Delta SP500$ is the S&P 500 index return and ΔCRB is the CRB Spot Index return, $\Delta FXVol$ is the currency volatility, and ΔLIQ is the liquidity, as proxied by the TED spread. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	CTLong	*CTShort*	*FundLong*	*FundShort*	*MomLong*	*MomShort*	$\Delta SP500$	ΔCRB	$\Delta FXVol$	ΔLIQ
CTLong	1									
CTShort	−0.61***	1								
FundLong	0.65***	−0.71***	1							
FundShort	−0.72***	0.80***	−0.58***	1						
MomLong	0.84***	−0.66***	0.77***	−0.65***	1					
MomShort	−0.73***	0.68***	−0.59***	0.83***	−0.56***	1				
$\Delta SP500$	0.23***	0.05	0.02	−0.15*	0.15**	−0.10	1			
ΔCRB	0.34***	−0.15**	0.15**	−0.34***	0.24***	−0.27**	0.39***	1		
$\Delta FXVol$	−0.29***	0.03	−0.05	0.16**	−0.22***	0.17**	−0.40***	−0.32***	1	
ΔLIQ	−0.27***	−0.03	−0.05	0.14**	−0.21***	0.11	−0.36***	−0.35***	0.28***	1

predictors $\Delta SP500_{t-1}$ and ΔCRB_{t-1} as vector $\mathbf{x_t}$ and subsequently include every variable of vector $\mathbf{x_t}$. Similar to Bakshi and Panayotov (2013) we use Newey and West (1987) corrected standard errors and select the lags using the Akaike information criterion.

Table 4 reports the slope coefficients of the univariate versions of Eq. (21). The coefficient for $\Delta SP500_{t-1}$ predicting *CTShort* is statistically significant, with a value of 0.154, indicating that the S&P 500 index return predicts the short leg of the carry trade strategy. The term $\Delta SP500_{t-1}$ is a statistically significant predictor of *FundLong* (with a coefficient of −0.112). For the CRB Spot Index, we find that ΔCRB_{t-1}

Table 4
Univariate predictability of FX strategy returns.

	CTLong	*CTShort*	*FundLong*	*FundShort*	*MomLong*	*MomShort*
$\Delta SP500_{t-1}$	−0.020	0.154**	−0.112**	0.023	−0.054	0.048
	[−0.204]	[2.164]	[−2.496]	[0.277]	[−0.993]	[0.616]
N	227	227	227	227	216	216
Adjusted R^2	−0.004	0.021	0.012	−0.004	−0.002	−0.003
	CTLong	*CTShort*	*FundLong*	*FundShort*	*MomLong*	*MomShort*
ΔCRB_{t-1}	0.292***	−0.007	0.047	−0.225***	0.103	−0.243***
	[3.579]	[−0.055]	[0.506]	[−3.206]	[1.400]	[−4.053]
N	227	227	227	227	216	216
Adjusted R^2	0.034	−0.004	−0.003	0.028	0.002	0.031

This table presents the univariate regression results of the in-sample predictability of the long and short legs of three currency strategies. The regression equation is $y_t = \delta_0 + \mathbf{x_{t-1}}'\boldsymbol{\delta_\chi} + \epsilon_t$, where y_t is the monthly return of *CTLong, CTShort, FundLong, FundShort, MomLong,* and *MomShort*. The variables *CTLong* and *CTShort* are the long and short legs, respectively, of a carry trade strategy buying currencies with high interest rates and selling currencies with low interest rates in relation to the USD. The variables *FundLong* and *FundShort* are the long and short legs, respectively, of a fundamental strategy buying undervalued currencies and selling overvalued currencies in relation to the three-year average real exchange rate. The variables *MomLong* and *MomShort* are the long and short legs, respectively, of a momentum strategy with a 12-month formation period and a one-month skip period. $\boldsymbol{\delta} = [\delta_0, \boldsymbol{\delta_\chi}']'$ is the vector of the regression coefficients. The explanatory variable $\mathbf{x_{t-1}}$ is either $\Delta SP500_{t-1}$ or ΔCRB_{t-1}, which are the three-month normalized predictors. The variable $\Delta SP500_{t-1}$ is the S&P 500 index return and ΔCRB_{t-1} is the CRB Spot Index return. Intercepts are calculated but not reported, for brevity. We use Newey–West (1987) corrected standard errors and the *t*-statistics are in brackets; ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

predicts the long leg return of the carry trade (with a coefficient of 0.292). These results for the long and short legs of the carry trade are consistent with the studies of Bakshi and Panayotov (2013) and Lu and Jacobsen (2016). We also report the significant predictive ability of ΔCRB_{t-1} on *FundShort* (−0.225) and *MomShort* (−0.243). The adjusted R-squared of ΔCRB_{t-1} predicting *CTLong* (3.4%) and that of $\Delta SP500_{t-1}$ predicting *CTShort* (2.1%) are similar to those reported by Bakshi and Panayotov (2013) and Lu and Jacobsen (2016). The adjusted R-squared of the other strategies with significant predictors are of similar magnitude. Lu and Jacobsen (2016) document that the equity market has significant ability to predict the short leg of the carry trade, and Bakshi and Panayotov (2013) also show that the CRB Spot Index predicts the long leg returns of the carry trade. The finding that an increase in $\Delta SP500_{t-1}$ is associated with lower future returns for undervalued currencies, *FundLong*, is novel to the literature. The results that the CRB Spot Index return predicts the short leg return of the fundamental and momentum strategies are also novel to the literature. Taken together, the results in Table 4 are consistent with Hong and Stein's (1999) gradual information diffusion hypothesis.

In Table 5, we estimate the multivariate versions of Eq. (21), including the control variables $\Delta FXVol_{t-1}$ and ΔLIQ_{t-1}. Compared to the univariate analysis results, we find a general increase in the significance of the predictors and the adjusted R-squareds in the estimated regressions models. It is noteworthy that, after we control for FX volatility, $\Delta FXVol_{t-1}$, and funding liquidity, ΔLIQ_{t-1}, in the multivariate analysis, the $\Delta SP500_{t-1}$ continues to be a strong predictor for several FX strategies.[14] More specifically, we find that $\Delta SP500_{t-1}$ negatively predicts the long leg return of the carry trade strategy (with a statistically significant coefficient of −0.165) and that the predictive ability of $\Delta SP500_{t-1}$ on *CTShort* remains, with only a marginal change in the coefficient, from 0.154 to 0.168, compared to the univariate regressions. The results also remain mostly unchanged in predicting *FundLong*; the coefficient for $\Delta SP500_{t-1}$ remains statistically significant and changes from −0.112 to −0.120. Moreover, $\Delta SP500_{t-1}$ has a weak ability to predict *FundShort* (0.117) and *MomShort* (0.143). Compared with previous studies, these coefficients and the adjusted R-squared are largely similar to those presented by Lu and Jacobsen (2016). These findings imply that the information diffusion between the U.S. equity market and the FX market is gradual and that the predictability is not altered by the inclusion of the control factors related to the risk premia of Bakshi and Panayotov (2013).

With regard to commodity returns as a predictor of FX returns, the results from the multivariate analysis reported in Table 5 remain qualitatively the same as the univariate analysis results in Table 4. The most noteworthy change is the decrease in the coefficient value for ΔCRB_{t-1}, from 0.292 to 0.188, in predicting *CTLong*. These results and the corresponding adjusted R-squared are still similar to the results of Bakshi and Panayotov (2013) and Lu and Jacobsen (2016), who find that the CRB Spot Index return predicts only the returns of the long leg of the carry trade and the combined carry trade strategy. The multivariate analysis results reported in Table 5 also support the gradual diffusion of information from the equity and commodity markets to the FX market.

3.2. Speculator activity and FX return predictability

To examine our hypothesis that the predictive effect of the equity and commodity market returns on FX strategy returns is weaker when FX speculators are active, we estimate regressions of the following form:

$$y_t = d_0 + \gamma_0 + \mathbf{x'_{t-1}}\boldsymbol{\gamma_\chi} + \eta_t, \tag{22}$$

where $y_t = \mathrm{r}_t^{\text{Strategy, Leg}}$ is the return of an FX strategy, $\gamma = [\gamma_0, \gamma_\chi']'$ is the vector of regression coefficients, $d_0 \mathrm{d}_0$ is a dummy variable for speculator activity, and the set of predictors and controls variables $\mathbf{x_{t-1}}$ is either $\mathbf{x_{t-1}} = [\Delta SP500_{t-1}, D_{t-1} \cdot \Delta SP500_{t-1}, \Delta FXVol_{t-1}, \Delta LIQ_{t-1}]$ or $\mathbf{x_{t-1}} = [\Delta \mathrm{CRB}_{t-1}, D_{t-1} \cdot \Delta CRB_{t-1}, \Delta FXVol_{t-1}, \Delta LIQ_{t-1}]$. We estimate Eq. (22) with and without the control variables $\Delta FXVol_{t-1}$ and ΔLIQ_{t-1}.

The variables of interest in Eq. (22) are the interaction terms involving the equity and commodity market returns, as well as the speculators' state of activity, which indicates whether there is a significant difference in predictability for greater speculation activity. Furthermore, we want to explicitly address whether there is predictability when speculators are active, not just whether there is a difference. Therefore, we test the following hypothesis:

[14] Lu and Jacobsen (2016) also show that equity returns predict the short leg of the carry trade. However, they do not use a multivariate setting to control for this. Instead, they use a set of other predictors as univariate replacements for the equity predictors and conclude that none of these can predict the short leg of the carry trade.

Table 5
Multivariate analysis of FX return predictability.

	CTLong	*CTShort*	*FundLong*	*FundShort*	*MomLong*	*MomShort*
$\Delta SP500_{t-1}$	−0.165**	0.168**	−0.120*	0.117*	−0.098	0.143*
	[−2.319]	[2.260]	[−1.815]	[1.703]	[−1.382]	[1.869]
$\Delta FXVol_{t-1}$	−0.054**	0.019	−0.005	0.037*	−0.006	0.036
	[−2.098]	[0.988]	[−0.283]	[1.827]	[−0.304]	[1.515]
ΔLIQ_{t-1}	−0.015	−0.002	−0.0002	0.009	−0.006	0.008
	[−1.642]	[−0.478]	[−0.055]	[1.065]	[−0.945]	[0.981]
Constant	1.000**	0.117	0.035	−0.421	0.436	−0.354
	[2.418]	[0.451]	[0.134]	[−1.014]	[1.147]	[−0.922]
N	227	227	227	227	216	216
Adjusted R^2	0.051	0.016	0.004	0.024	−0.002	0.016
	CTLong	*CTShort*	*FundLong*	*FundShort*	*MomLong*	*MomShort*
	CTLong	*CTShort*	*FundLong*	*FundShort*	*MomLong*	*MomShort*
ΔCRB_{t-1}	0.188**	−0.039	0.079	−0.168**	0.093	−0.211***
	[2.073]	[−0.380]	[0.996]	[−2.068]	[1.002]	[−2.671]
$\Delta FXVol_{t-1}$	−0.029	0.002	0.010	0.017	0.009	0.011
	[−1.220]	[0.157]	[0.661]	[0.925]	[0.453]	[0.519]
ΔLIQ_{t-1}	−0.010	−0.005	0.003	0.005	−0.003	0.003
	[−1.085]	[−1.299]	[0.729]	[0.638]	[−0.629]	[0.329]
Constant	0.628	0.361	−0.184	−0.126	0.240	−0.003
	[1.525]	[1.455]	[−0.701]	[−0.342]	[0.653]	[−0.007]
N	227	227	227	227	216	216
Adjusted R^2	0.048	−0.007	−0.007	0.028	−0.005	0.024

This table presents the multivariate regression results of the in-sample predictability of the long and short legs of three currency strategies. The regression equation is $yt = \delta_0 + x'_{t-1}\delta_\chi + \epsilon t$, where yt is the monthly return of *CTLong, CTShort, FundLong, FundShort, MomLong,* and *MomShort*. The variables *CTLong* and *CTShort* are the long and short legs, respectively, of a carry trade strategy buying currencies with high interest rates and selling currencies with low interest rates in relation to the USD. The variables *FundLong* and *FundShort* are the long and short legs, respectively, of a fundamental strategy buying undervalued currencies and selling overvalued currencies in relation to the three-year average real exchange rate. The variables *MomLong* and *MomShort* are the long and short legs, respectively, of a momentum strategy with a 12-month formation period and a one-month skip period. $\delta = [\delta_0, \delta'_\chi]'$ is the vector of the regression coefficients, and either $x_{t-1} = [\Delta SP500_{t-1}, \Delta FXVol_{t-1}, \Delta LIQ_{t-1}]$ or $x_{t-1} = [\Delta CRB_{t-1}, \Delta FXVol_{t-1}, \Delta LIQ_{t-1}]$, which are the three-month normalized predictors and controls. The variable $\Delta SP500t-1$ is the S&P 500 index return and ΔCRB_{t-1} is the CRB Spot Index return, $\Delta FXVol_{t-1}$ is currency volatility, and ΔLIQ_{t-1} is the liquidity, as proxied by the TED spread. We use Newey–West (1987) corrected standard errors and the t-statistics are in brackets; ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

$$H_0: \gamma_{Pred} + \gamma_{interacted} = 0, \tag{23}$$

where γ_{Pred} is the coefficient of the predictor of interest and $\gamma_{interacted}$ is the coefficient of the interaction term.[15] We thus test whether there is predictability when speculators are active and report the corresponding *t*-statistic.

Table 6 reports the results of the speculator activity effect on the predictability of FX strategy returns. For the $\Delta SP500_{t-1}$ predictor, the coefficient of the interaction term, $D_{t-1} \cdot \Delta SP500_{t-1}$, enables us to identify a statistically significant difference in the predictive ability of the S&P 500 index when speculators are active for the short leg of the carry trade strategy, *CTShort*, and the long leg of the fundamental strategy, *FundLong*. The results in Panel A reveal statistically significant coefficients of −0.295 and 0.158 for *CTShort* and *FundLong*, respectively (see the second and third columns). Both of these coefficients have the opposite sign of the coefficient of $\Delta SP500_{t-1}$ but are similar in magnitude (0.223 and −0.157, respectively), indicating a mitigating effect on predictability. These results support our hypothesis and suggest that speculator activity has a significant and measurable influence on cross-asset information diffusion.

The *t*-statistic and *p*-value from the hypothesis test in Eq. (22), reveal that the sum of the coefficients of $\Delta SP500_{t-1}$ and $D_{t-1} \cdot \Delta SP500_{t-1}$ is not significantly different from zero for any FX strategy. This result indicates that the S&P 500 index cannot predict any FX strategies when speculators are active. We therefore conclude that, when speculators are active, the predictive ability of equity market returns on FX market strategies disappears. This finding is in line with our hypothesis and suggests that speculator activity increases the rate of information diffusion across assets.

The results from assessing the predictability of commodity market returns, the CRB Spot Index returns, conditioned on speculator activity are reported in Panel B of Table 6. The coefficient of the interaction term $D_{t-1} \cdot \Delta CRB_{t-1}$ is statistically nonsignificant for every FX strategy, lending no support for the hypothesis that the ability of commodity market returns to predict FX strategy returns is weaker when FX speculators are active. However, the hypothesis test results for the sum of the coefficients for ΔCRB_{t-1} and $D_{t-1} \cdot \Delta CRB_{t-1}$ in Panel B are nonsignificant for all FX strategies, except for *MomShort*, indicating that, when speculators are active, the predictability of both *CTLong* and *FundShort* dissipates, supporting our central hypothesis that the ability of the equity and commodity market returns to predict FX strategy returns is weaker when FX speculators are active.

Table 7 presents the results of the multivariate versions of Eqs. (22) and (23) with the control variables $\Delta FXVol_{t-1}$ and ΔLIQ_{t-1}. In Panel A, the coefficients for the predictor $\Delta SP500_{t-1}$ are larger and they are significant in a larger number of cases compared to those in the multivariate analysis without speculator activity in Table 5. The interaction term $D_{t-1} \cdot \Delta SP500_{t-1}$ consistently has the opposite sign compared to the coefficient of the predictor $\Delta SP500_{t-1}$, indicating a mitigating effect on predictability when speculators are active. For example, the

[15] If the standard error of either γ_{Pred} or $\gamma_{interacted}$ is large, it will lead to the standard error of the sum of the coefficients being large which could bias the results in favour of our hypothesis. To address this concern, we change the definition of the dummy variable to $D'_t = D_t - 1$, that is equal to one when speculators are inactive (instead of active) and re-run the regressions. We then test our main hypothesis that there is no predictability when speculators are active with a *t*-test of the predictor coefficient. Our results are identical with this modified approach suggesting our results do not suffer from the bias.

Table 6
Investor activity and the univariate predictability of FX strategy returns

Panel A: $\Delta SP500_{t-1}$ predictor

	CTLong	*CTShort*	*FundLong*	*FundShort*	*MomLong*	*MomShort*
$\Delta SP500_{t-1}$	−0.014	0.223***	−0.157***	0.024	−0.066	0.082
	[−0.132]	[3.127]	[−2.613]	[0.253]	[−0.959]	[0.935]
$D_{t-1} \cdot \Delta SP500_{t-1}$	−0.023	−0.295**	0.158*	−0.064	0.048	−0.120
	[−0.143]	[−2.288]	[1.681]	[−0.460]	[0.398]	[−0.995]
D_{t-1}	0.034	0.637**	−0.056	0.608*	−0.096	0.091
	[0.079]	[2.143]	[−0.177]	[1.670]	[−0.247]	[0.261]
Constant	0.174	−0.172	0.001	−0.161	0.144	0.074
	[0.565]	[−0.751]	[0.004]	[−0.533]	[0.453]	[0.271]
N	227	227	227	227	216	216
Adjusted R^2	−0.013	0.037	0.010	−0.002	−0.011	−0.010
$\boldsymbol{H_0}: \gamma_1 \Delta SP500_{t-1} + \gamma_2 D_{t-1} \cdot \Delta SP500_{t-1} = 0$						
t-Stat	0.084	0.669	0.010	0.333	0.162	0.356
p-Value	0.791	0.504	0.994	0.739	0.872	0.722

Panel B: ΔCRB_{t-1} predictor

	CTLong	*CTShort*	*FundLong*	*FundShort*	*MomLong*	*MomShort*
ΔCRB_{t-1}	0.331***	−0.027	0.020	−0.280***	0.142	−0.242***
	[3.829]	[−0.178]	[0.178]	[−3.368]	[1.308]	[−3.719]
$D_{t-1} \cdot \Delta CRB_{t-1}$	−0.122	−0.010	0.125	0.103	−0.125	−0.045
	[−0.661]	[−0.052]	[0.847]	[0.823]	[−0.687]	[−0.422]
D_{t-1}	−0.183	0.585**	−0.130	0.726**	−0.166	0.283
	[−0.475]	[2.064]	[0.310]	[2.475]	[−0.514]	[0.321]
Constant	0.242	−0.165	−0.004	−0.218	0.188	0.003
	[0.901]	[−0.717]	[−0.020]	[−0.895]	[0.676]	[0.013]
N	227	227	227	227	216	216
Adjusted R^2	0.034	−0.004	−0.003	0.028	0.002	0.031
$\boldsymbol{H_0}: \gamma_1 \Delta CRB_{t-1} + \gamma_2 D_{t-1} \cdot \Delta CRB_{t-1} = 0$						
t-Stat	1.311	0.320	1.610	1.606	0.133	2.496
p-Value	0.191	0.473	0.109	0.110	0.895	0.013**

This table presents the univariate regression results of the in-sample predictability of the long and short legs of three currency strategies. The regression equation is $y_t = d_0 + \gamma_0 + \mathbf{x_{t-1}}'\gamma_\chi + \eta_t$, where y_t is the monthly return of *CTLong, CTShort, FundLong, FundShort, MomLong,* and *MomShort.* The variables *CTLong* and *CTShort* are the long and short legs, respectively, of a carry trade strategy buying currencies with high interest rates and selling currencies with low interest rates in relation to the USD. The variables *FundLong* and *FundShort* are the long and short legs, respectively, of a fundamental strategy buying undervalued currencies and selling overvalued currencies in relation to the three-year average real exchange rate. The variables *MomLong* and *MomShort* are the long and short legs, respectively, of a momentum strategy with a 12-month formation period and a one-month skip period. $\gamma = [\gamma_0, \gamma_\chi']'$ is the vector of the regression coefficients, and in Panel A $\mathbf{x_{t-1}} = [\Delta SP500_{t-1}, D_{t-1} \cdot \Delta SP500_{t-1}]$, and in Panel B $\mathbf{x_{t-1}} = [\Delta CRB_{t-1}, D_{t-1} \cdot \Delta \mathrm{CRB}_{t-1}]$, which are the three-month normalized predictors and a dummy variable. The variable $\Delta SP500_{t-1}$ is the S&P 500 index return and ΔCRB_{t-1} is the CRB Spot Index and D_{t-1} is a dummy variable indicating high speculator activity. The table also presents the *t*-statistic and *p*-value from the hypothesis test $H_0: \gamma_{Pred} + \gamma_{interacted} = 0$, where γ_{Pred} is the coefficient of a predictor and $\gamma_{interacted}$ is the coefficient of the interaction term. We use Newey–West (1987) corrected standard errors and the *t*-statistics are in brackets; ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

coefficient for the interaction term $D_{t-1} \cdot \Delta SP500_{t-1}$ for *CTShort* is negative (−0.321) and statistically significant at the 5% level, suggesting a significant difference in predictability when speculators are active. In line with the results in Table 6, the test statistics in the lower part of the table for the hypothesis that the sum of $\Delta SP500_{t-1}$ and $D_{t-1} \cdot \Delta SP500_{t-1}$ is equal to zero suggest no predictability when speculators are active. The results in Table 7 thus continue to show evidence that the ability of the equity market to predict the FX market is conditional on speculator activity, supporting our central hypothesis and suggesting that speculator activity increases the rate of information diffusion in currency strategies.

The results for commodity market returns, reported in Panel B of Table 7, reveal statistically significant coefficients of ΔCRB_{t-1} for predicting *CTLong*, *FundShort*, and *MomShort*, with coefficient values of 0.205, −0.203, and − 0.198, respectively. As in previous tables, this result indicates that commodity market returns have the ability to predict FX strategy returns. However, the results assessing the information diffusion from the commodity market to the FX market provide only partial support for our hypothesis that the ability of the equity and commodity market returns to predict FX strategy returns is weaker when FX speculators are active. The coefficients for the interaction term $D_{t-1} \cdot \Delta CRB_{t-1}$ are not statistically significant. In addition, the results for the hypothesis test of the sum of the coefficients of ΔCRB_{t-1} and $D_{t-1} \cdot \Delta CRB_{t-1}$ indicate the predictive ability of *MomShort* and *FundLong* when speculators are active, whereas this ability has dissipated for *CTLong* and *FundShort*, partly supporting our hypothesis. Generally, the results in Table 7 suggest that speculator activity is a relevant determinant of information diffusion across markets.

Overall, our results show that cross-asset market predictability from the equity and commodity markets to the FX market depends on speculators actively investing in the carry trade strategy. These results are consistent with studies on the equity market showing evidence that the availability of information is positively related to the rate of its diffusion. For instance, Hong et al. (2000) find that momentum in the

Table 7
Investor activity and the multivariate predictability of FX strategy returns.

Panel A: $\Delta SP500_{t-1}$ predictor

	CTLong	*CTShort*	*FundLong*	*FundShort*	*MomLong*	*MomShort*
$\Delta SP500_{t-1}$	−0.196***	0.262***	−0.174**	0.156**	−0.129	0.217**
	[−2.639]	[3.113]	[−2.102]	[2.055]	[−1.579]	[2.376]
$D_{t-1} \cdot \Delta SP500_{t-1}$	0.109	−0.321**	0.170	−0.159	0.098	−0.223
	[0.822]	[−2.280]	[1.584]	[−1.367]	[0.802]	[−1.640]
$\Delta FXVol_{t-1}$	−0.058**	0.029	−0.008	0.045**	−0.009	0.042*
	[−2.260]	[1.403]	[−0.382]	[2.265]	[−0.442]	[1.757]
ΔLIQ_{t-1}	−0.016	−0.0005	−0.001	0.011	−0.007	0.009
	[−1.646]	[−0.108]	[−0.221]	[1.137]	[−0.983]	[1.063]
D_{t-1}	−0.310	0.731**	−0.091	0.861**	−0.167	0.271
	[−0.814]	[2.299]	[−0.283]	[2.404]	[−0.451]	[0.742]
Constant	1.139**	−0.203	0.061	−0.822*	0.507	−0.464
	[2.340]	[−0.633]	[0.202]	[−1.699]	[1.143]	[−1.073]
N	227	227	227	227	216	216
Adjusted R^2	0.045	0.037	0.002	0.038	−0.010	0.016
$\boldsymbol{H_0}: \gamma_1 \Delta SP500_{t-1} + \gamma_2 D_{t-1} \cdot \Delta SP500_{t-1} = 0$						
t-Stat	0.683	0.521	0.045	0.032	0.295	0.052
p-Value	0.494	0.603	0.965	0.975	0.769	0.959

Panel B: ΔCRB_{t-1} predictor

	CTLong	*CTShort*	*FundLong*	*FundShort*	*MomLong*	*MomShort*
ΔCRB_{t-1}	0.205**	−0.059	0.052	−0.203**	0.129	−0.198*
	[2.022]	[−0.433]	[0.528]	[−2.023]	[1.053]	[−1.971]
$D_{t-1} \cdot CRB_{t-1}$	−0.029	0.020	0.101	0.047	−0.111	−0.077
	[−0.151]	[0.118]	[0.787]	[0.328]	[−0.636]	[−0.504]
$\Delta FXVol_{t-1}$	−0.032	0.007	0.009	0.024	0.008	0.013
	[−1.312]	[0.443]	[0.605]	[1.326]	[0.411]	[0.661]
ΔLIQ_{t-1}	−0.010	−0.005	0.002	0.005	−0.003	0.003
	[−1.102]	[−1.134]	[0.622]	[0.648]	[−0.553]	[0.367]
D_{t-1}	−0.360	0.567**	−0.083	0.842***	−0.163	0.340
	[−1.011]	[2.011]	[−0.268]	[2.807]	[−0.594]	[1.010]
Constant	0.802*	0.088	−0.148	−0.533	0.325	−0.165
	[1.870]	[0.290]	[−0.486]	[−1.303]	[0.843]	[−0.418]
N	227	227	227	227	216	216
Adjusted R^2	0.043	−0.005	−0.015	0.043	−0.012	0.018
$\boldsymbol{H_0}: \gamma_1 \Delta CRB_{t-1} + \gamma_2 D_{t-1} \cdot \Delta CRB_{t-1} = 0$						
t-Stat	1.128	0.359	1.775	1.377	0.136	2.366
p-Value	0.261	0.721	0.077*	0.173	0.892	0.019**

This table presents the univariate regression results of the in-sample predictability of the long and short legs of three currency strategies. The regression equation is $y_t = d_0 + \gamma_0 + \mathbf{x_{t-1}}'\gamma_\chi + \eta_t$, where y_t is the monthly return of *CTLong, CTShort, FundLong, FundShort, MomLong,* and *MomShort*. The variables *CTLong* and *CTShort* are the long and short legs, respectively, of a carry trade strategy buying currencies with high interest rates and selling currencies with low interest rates in relation to the USD. The variables *FundLong* and *FundShort* are the long and short legs, respectively, of a fundamental strategy buying undervalued currencies and selling overvalued currencies in relation to the three-year average real exchange rate. The variables *MomLong* and *MomShort* are the long and short legs, respectively, of a momentum strategy with a 12-month formation period and a one-month skip period. $\gamma = [\gamma_0, \gamma_\chi']'$ is the vector of the regression coefficients, and in Panel A $\mathbf{x_{t-1}} = [\Delta SP500_{t-1}, D_{t-1} \cdot \Delta SP500_{t-1}, \Delta FXVol_{t-1}, \Delta LIQ_{t-1}]$, and in Panel B $\mathbf{x_{t-1}} = [\Delta \mathrm{CRB}_{t-1}, D_{t-1} \cdot \Delta CRB_{t-1}, \Delta FXVol_{t-1}, \Delta LIQ_{t-1}]$ in Panel B, which are the three-month normalized predictors, controls, and a dummy variable. The variable $\Delta SP500_{t-1}$ is the S&P 500 index return and ΔCRB_{t-1} is the CRB Spot Index return, $\Delta FXVol_{t-1}$ is currency volatility, ΔLIQ_{t-1} is the liquidity as proxied by the TED spread, and D_{t-1} is a dummy variable for high speculator activity. We use Newey and West (1987) corrected standard errors and the t-statistics are in brackets; ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

equity market is stronger for stocks with low analyst coverage, and Chordia and Swaminathan (2000) find that stocks with high trading volumes lead stocks with low trading volumes. Further, Menzly and Ozbas (2010) present direct evidence that the magnitude of return cross-predictability within the equity market declines with the number of informed investors, as proxied by analyst coverage and institutional ownership. Our finding that speculator activity reduces the predictability of FX returns corroborates these findings and extends them to the FX market.

3.3. Analysis of information diffusion

The finding that speculator activity does not affect the ability of the commodity market to predict the FX market to the same extent as it affects the equity market is puzzling. An explanation for this finding could, however, be that speculators in the FX market do not sufficiently monitor or exploit information from the commodity market. This could be related to the finding of Kohlscheen, Avalos, and Schrimpf (2017) of a distinct commodity-related predictor of exchange rate movement that is unrelated to global risk appetite. If speculators do not monitor or trade on this information, this could explain why their activity does not clearly improve information diffusion from the commodity market to the FX market. An alternative explanation could be related to the time-varying risk premium interpretation of carry trade returns of Bakshi and Panayotov (2013). Further, Ready, Roussanov, and Ward (2017) differentiate between commodity countries that export basic commodities and producer countries that import basic goods and export finished consumption goods. In line with the risk premium interpretation of carry trade returns, Ready et al. (2017) relate the predictability of carry trade returns to commodity currencies that earn a risk premium. Thus, if the observed predictability of FX returns is unrelated to information diffusion, it is logical that speculator activity, which should increase the rate of information diffusion between the markets, does not affect the predictability.

It is therefore possible that the ability of both the commodity and equity market returns to predictive FX strategy returns is not related to gradual information diffusion. We address this concern by evaluating whether investors react to information with a delay, as the gradual information diffusion hypothesis predicts. Driesprong et al. (2008) and Lu and Jacobsen (2016) argue that, if investors react to new information with a delay, the explanatory power of the predictive variables for FX strategy returns should peak after some lags and decrease relatively quickly thereafter. The authors test the gradual information diffusion hypothesis by plotting the R-squared values as a function of the lag length between the predictor and the returns of a strategy. In the spirit of Driesprong et al. (2008) and Lu and Jacobsen (2016), we use lag lengths of one week (five trading days) to 16 weeks and perform the analysis of Eq. (21) for each lag length.

Figs. 1 and 2 plot the R-squared as a function of lag length in weeks for the predictors $\Delta SP500_{t-1}$ and $\Delta \mathrm{CRB}_{t-1}$, respectively, for two subsamples: i) when speculator activity is high, that is, when the dummy variable is equal to one (dotted line), and ii) when speculator activity is low, that is, when the dummy variable is equal to zero (solid line). The patterns in the two figures support the gradual information diffusion explanation. First, for all the figures except for Figs. 1e and 2c, the explanatory power peaks after some lags (between five and 10 weeks) and decreases thereafter. Second, this pattern is almost exclusively present in periods with low speculator activity (solid line), lending credence to our hypothesis that speculator activity increases the rate of information diffusion. For instance, in Fig. 1b, where $\Delta SP500_{t-1}$ predicts *CTShort*, an R-squared value of nearly 10% is achieved for the fifth weekly lag in the subsample of low speculator activity, whereas the R-squared values for the subsample of high speculator activity never exceeds 4%. Comparing this result to those of Lu and Jacobsen (2016), who do not account for speculator activity, the peak explanatory power for the low-activity subsample is more than twice that presented in their study. Similarly, in Fig. 2a, where ΔCRB_{t-1} predicts *CTLong*, the R-squareds peak for the low-activity subsample is more than twice that reported by Lu and Jacobsen (2016).

3.4. Role of market frictions and illiquidity

Market frictions and illiquidity could affect speculators' activity and their ability to trade on information. Higher and lower levels of speculator activity could themselves be governed by limits to arbitrage and market frictions, forcing speculators to react more slowly or making them less willing to trade on information, which, in turn, gives rise to predictability. Indeed, the dummy variable has a statistically significant and negative correlation coefficient with $\Delta FXVol$ (−0.20) and ΔLIQ (−0.12), which suggests that speculator activity is lower when volatility is higher and funding constraints are tighter. However, our results are unaffected by the inclusion of these controls, as shown in the results of Tables 4 and 6.

To further address the role of market frictions and illiquidity, we include the three-month normalized innovations of the average bid–ask spread of Menkhoff et al. (2012) as a measure of market illiquidity as an additional control in Eq. (22). The results are available upon request. We find that our results remain qualitatively the same, suggesting that market frictions and illiquidity are not the primary reasons behind the importance of speculator activity in FX return predictability.

4. Further analysis

4.1. Alternative activity measures

As described in Section 2, our carry trade activity measure is based on end-of-month observations. For robustness, we re-estimate the regressions in Eqs. (21) and (22) using a carry trade activity measure based on the monthly averages of speculator positions. The results of the re-estimations are qualitatively the same. Since we measure speculator activity using positions in the AUD/JPY carry trade, we want to ensure that our results are not subject to the choice of these particular currencies. Therefore, we also re-estimate the analysis in Tables 5 to 7 using a broader measure of carry trade activity. We form the variable by taking an average of all the long positions in the AUD, CAD, and GBP minus the short positions in the JPY, CHF, and EUR, consistent with carry trade positions. The results using the broader measure of carry trade activity remain similar to those in our main analysis.

We also re-estimate Eqs. (21) and (22), using an alternative speculator activity measure based on the sentiment index created and used by Wang (2001, 2004). This alternative speculator activity measure represents the general activity of speculators and is based on past extreme values over the last three years. While Wang (2001, 2004) bases the sentiment index on net positions to obtain a bullish/bearish sentiment, we use total open interest to capture activity. Formally, the measure is

$$Activity_t = \frac{1}{N}\sum_{C}^{N}\frac{OI_t^C - Min(OI_t^C)}{Max(OI_t^C) - Min(OI_t^C)} \bullet 100, \tag{24}$$

where OI_t^C is the total open interest (long positions plus short positions) and $Max(OI_t^C)$ and $Min(OI_t^C)$ are, respectively, the maximum and minimum total open interest over the previous three-year period prior to week t.[16] The results obtained from the analysis using the alternative speculator measure based on Wang's (2001, 2004) are qualitatively similar, also leading to the conclusion that speculator activity diminishes predictability.

[16] The final measure is then standardized. Note also that the correlation of this measure with the carry trade activity measure is around 0.35.

4.2. Hedger activity

To ascertain that changes in FX return predictability are driven by speculator activity instead of broad currency market activity, we re-estimate the activity measure for the group commonly referred to as hedgers (labelled commercial traders by the CFTC). We then rerun the regressions in Eqs. (21) and (22) to evaluate whether there is a difference in predictability when hedgers are active in the carry trade strategy. We acknowledge that hedgers do not actively trade the carry trade strategy, although their positions may be consistent with it.

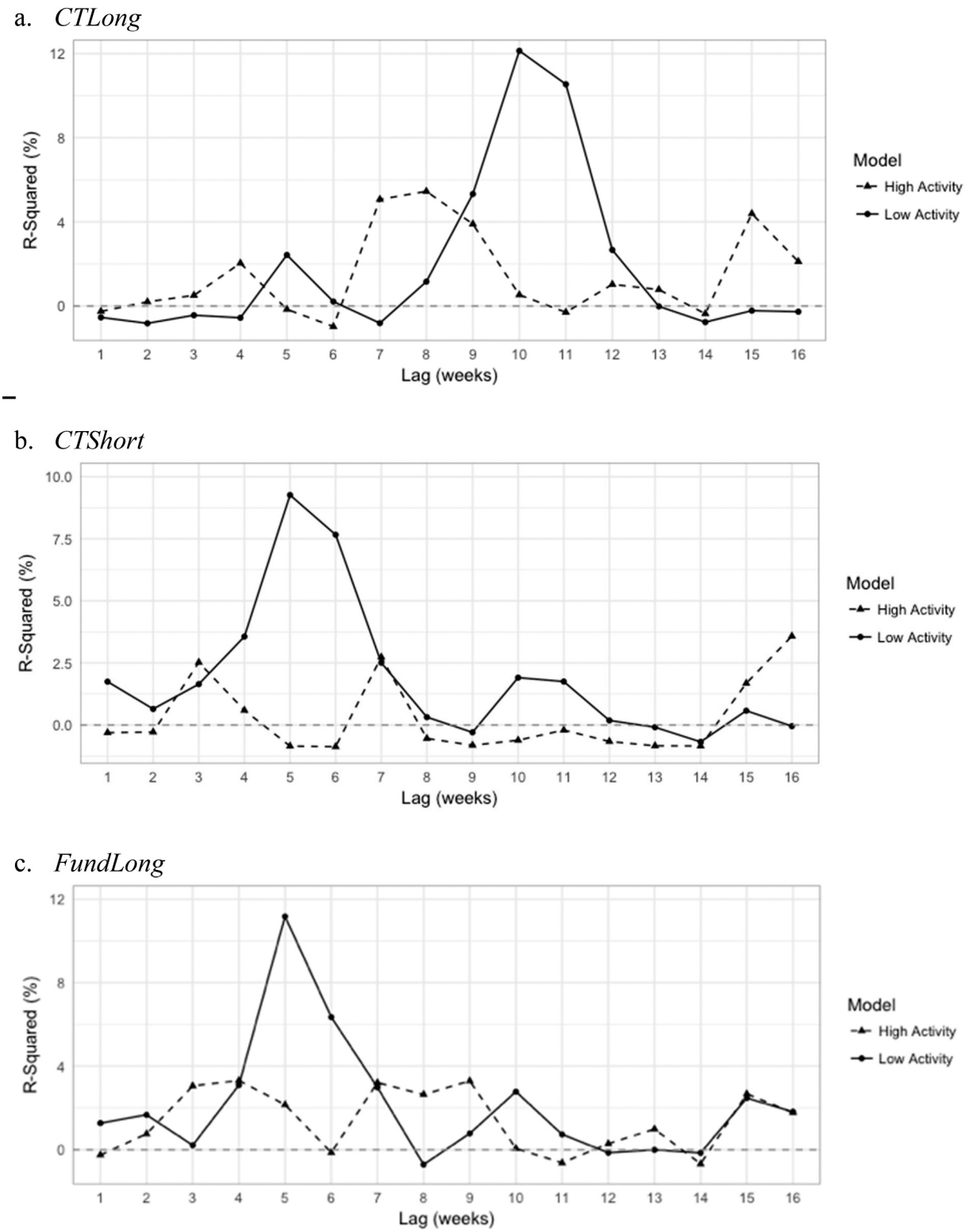
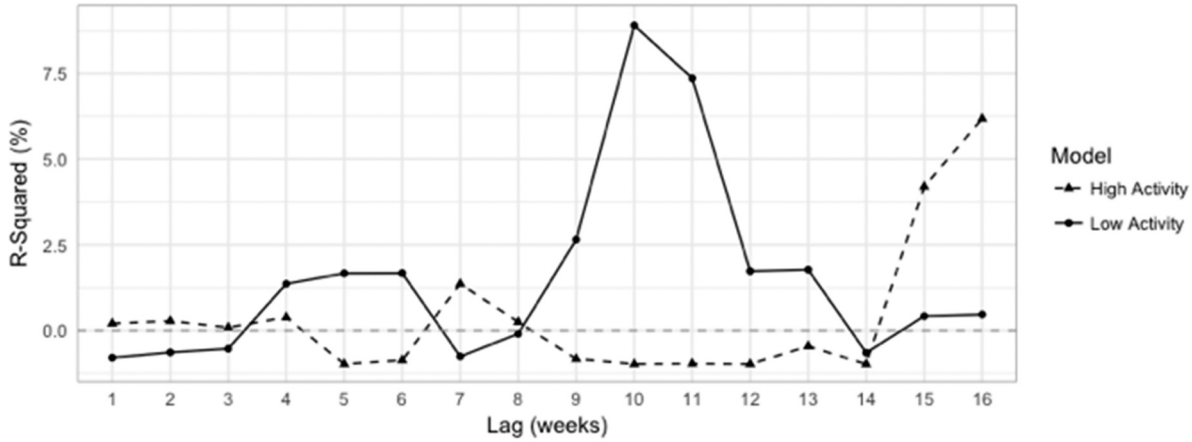
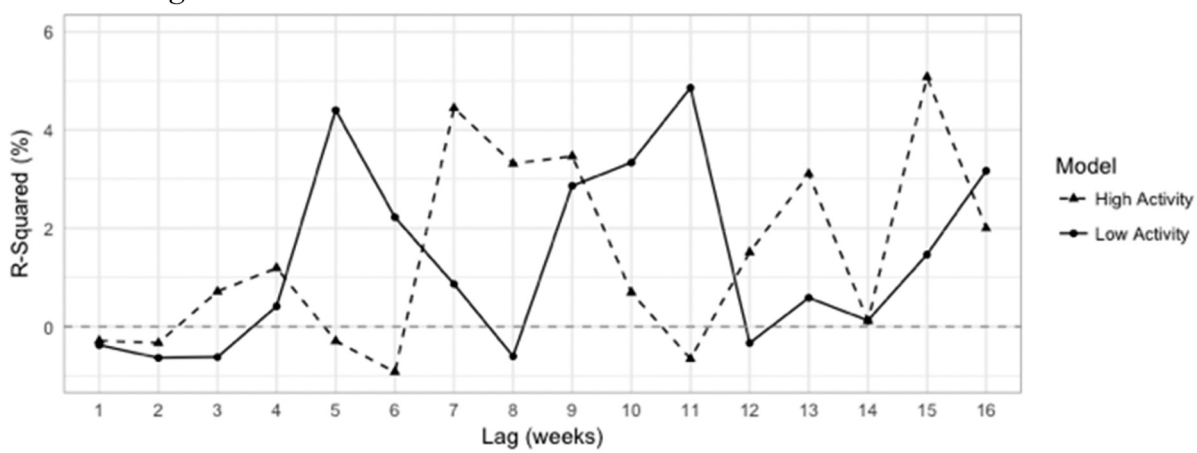


Fig. 1. Explanatory Power for Different Lag Lengths (Equity Market).
This figure plots R-squared as a function of the lag length in weeks for two subsamples; i) during high speculator activity (solid line) and ii) during low speculator activity (dashed line) for a univariate regression predicting the long and short legs of three currency strategies with S&P 500 equity index returns and controls.

d. *FundShort*



e. *MomLong*



f. *MomShort*

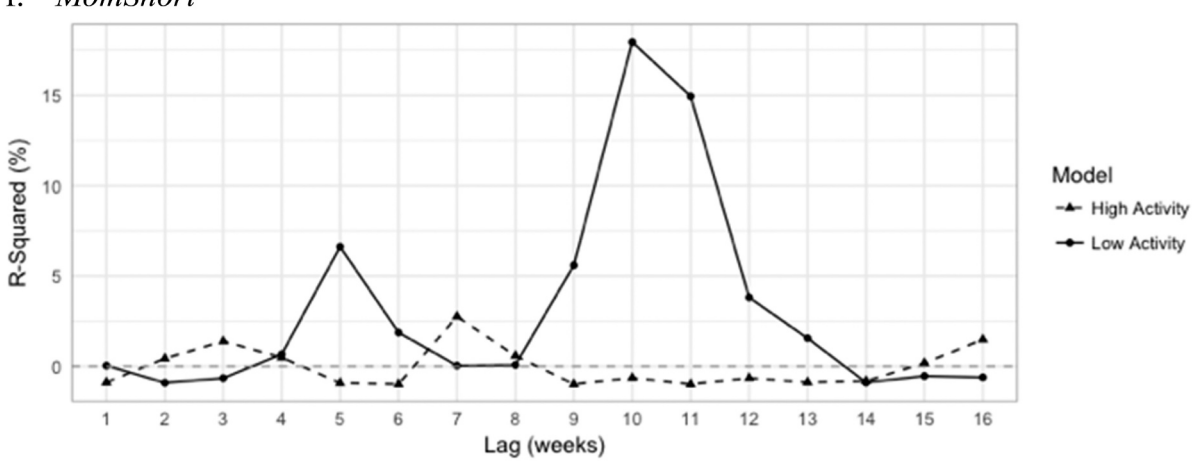


Fig. 1. (*continued*)

Hedgers can be considered uninformed investors who do not facilitate the information diffusion process, since they are trading to offset the risks of other positions. In untabulated results, we find that the ability of the equity and commodity markets to predict the FX strategies increases when hedgers are active. This finding confirms that it is particularly speculator activity that decreases the predictive ability of the equity market on the FX strategy returns. Considering hedgers as uninformed traders, the finding is in line with that of Menzly and Ozbas (2010), who relate cross-asset predictability to informed traders.

4.3. Information diffusion within assets

Given the significant correlations between some of the returns of the considered strategies' legs and the predictors (see Table 3), we re-estimate all the regressions in Eqs. (21) and (22), including a lagged dependent variable, to account for the possibility of omitted variable bias.

We do not tabulate the results of this analysis but conclude that this addition does not change our results. Furthermore, to account for the possibility of information diffusion within the strategies' own returns, so that the information contained in the lagged returns of a strategy's leg diffuses slowly into the next period's returns – which, in that case, could depend on speculator activity, as in our central hypothesis – we re-estimate all the regressions in Eqs. (21) and (22), including the lagged dependent variable and its interaction with the speculator activity dummy variable. This exercise marginally strengthens the results of this paper and does not support speculator activity having an effect on information diffusion within a strategy.

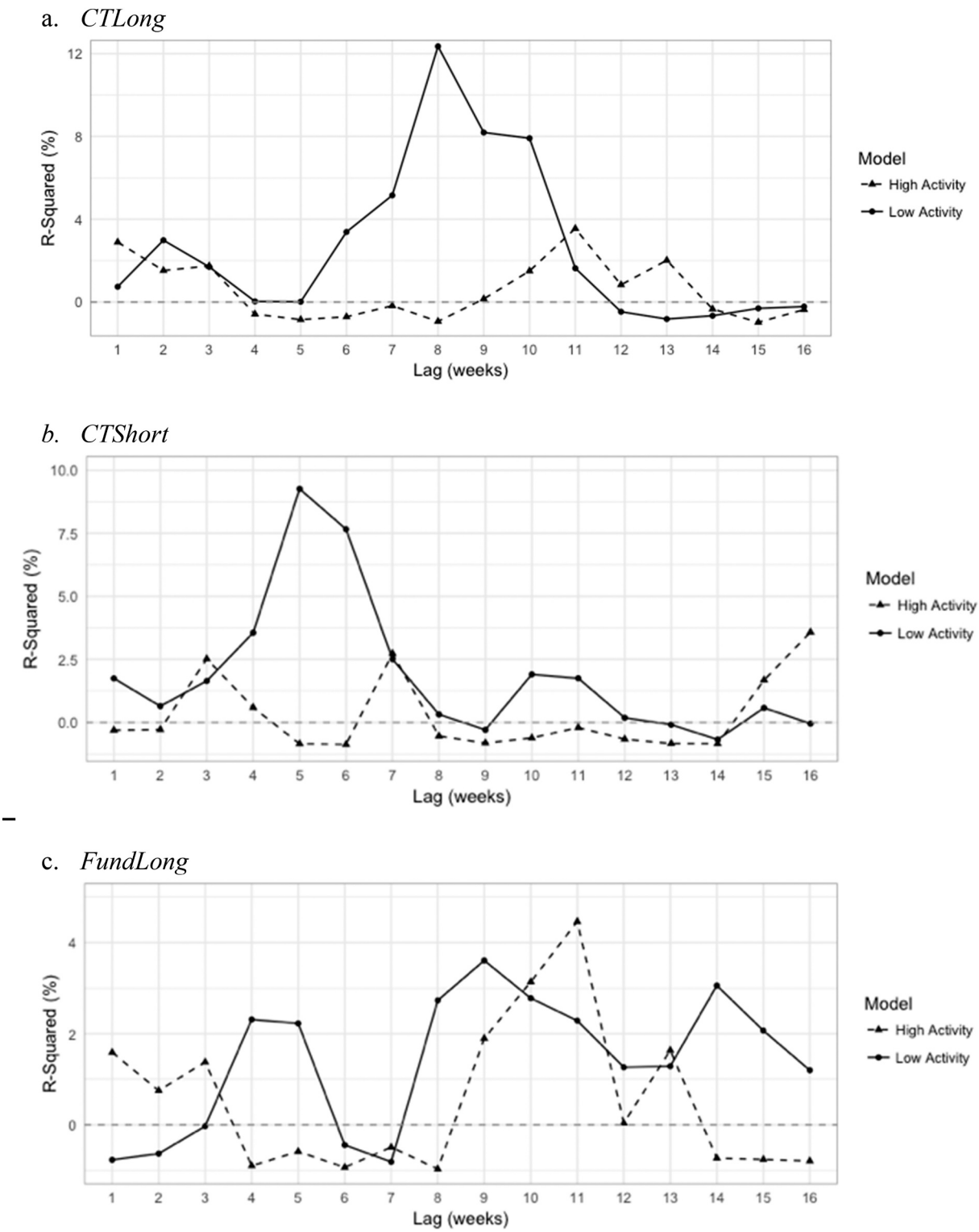


Fig. 2. Explanatory Power for Different Lag Lengths (Commodity Market).
This figure plots R-squared as a function of lag length in weeks for two subsamples; i) during high speculator activity (solid line) and ii) during low speculator activity (dashed line) for a univariate regression predicting the long and short legs of three currency strategies with CRB Spot Index returns and controls.

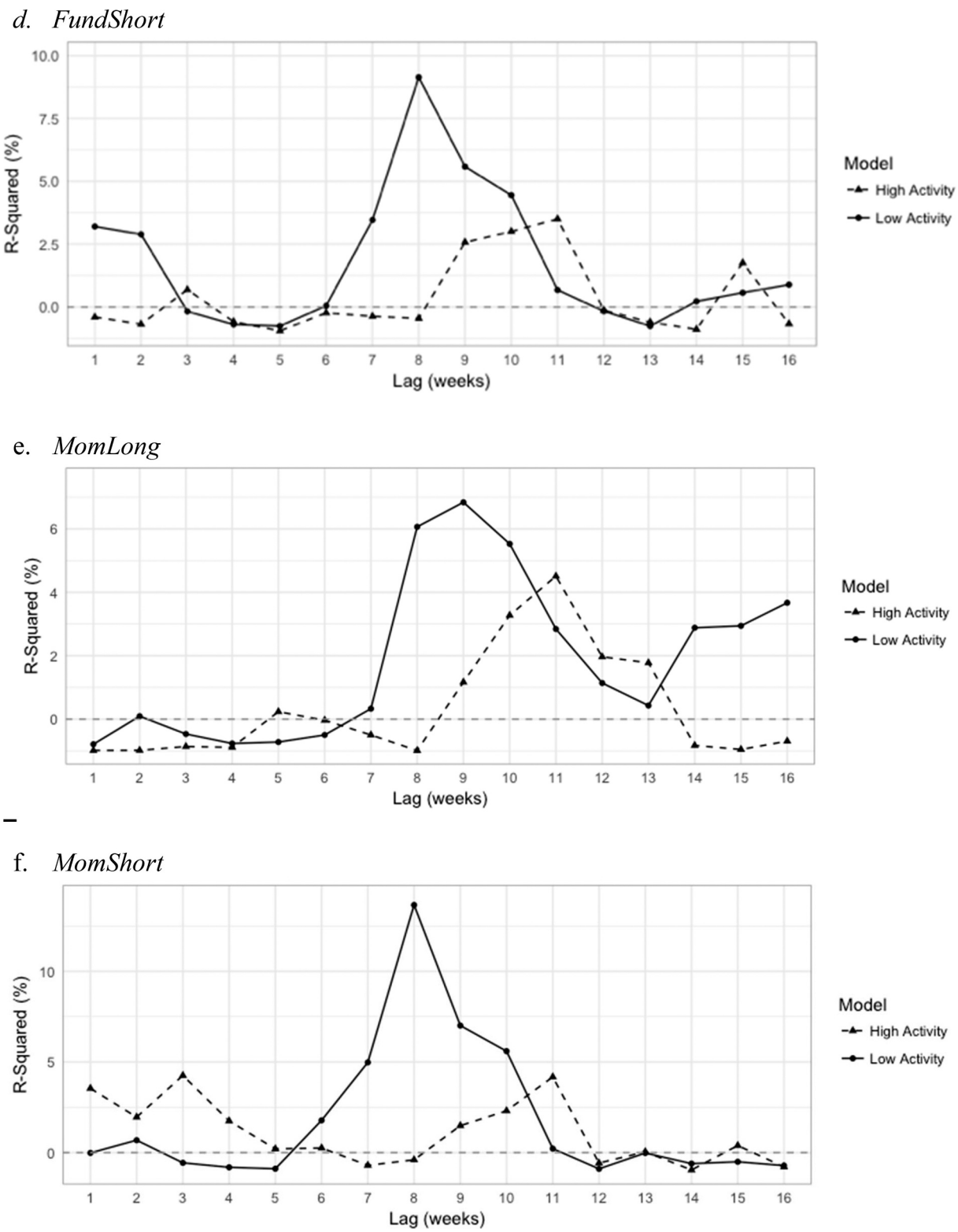


Fig. 2. (*continued*)

4.4. Additional robustness tests

Lu and Jacobsen (2016) consider the MSCI World Index as a predictor of the profits of the short leg of a carry trade. Therefore, we re-estimate Eqs. (21) and (22) using the MSCI World Index as our equity predictor. We find that using the MSCI World Index as the proxy for the equity market does not change our findings, although the predictive ability of the equity market on FX strategies slightly weakens with the MSCI World Index. It is noteworthy that the carry trade specification we are using is based on using the USD as the base currency. Therefore, the effect of the U.S. equity market could be more pronounced than the MSCI World Index. These results are available on request.

In addition to the inclusion of liquidity and FX volatility as control variables in Tables 5 to 7, we consider the Chicago Board Options Exchange (CBOE) Volatility Index (VIX). Brunnermeier et al. (2009), for example, find that the VIX is important in determining carry trade profits. Therefore, we obtain data for the VIX from the CBOE website and normalize relative changes in the index over three months, to be used as an additional control variable in Eqs. (21) and (22). In line with the results in Tables 5 to 7, we find (no) predictability when speculators are (active) inactive.

Next, we consider the possibility of speculator activity being an outcome of FX strategy returns. In untabulated analysis, we construct a dummy variable based on speculator activity that is unrelated to either FX strategy returns or the predictors $\Delta SP500$ and ΔCRB. That is, we build the dummy variable from the residuals of a regression of FX strategy returns and the predictors $\Delta SP500$ and ΔCRB on speculator activity. *Re*-estimating the regressions in Eqs. (21) and (22) with the orthogonalized speculator activity dummy does not change our results; if anything, it strengthens the results for $\Delta SP500$.

5. Conclusion

We test the gradual information diffusion hypothesis to evaluate the role of speculative activity in the FX market regarding information flow from the equity and commodity markets into FX strategies. The information flow has important implications for the cross-asset predictability of FX strategy returns, due to the inability of investors to process and monitor every piece of information. We argue that informed and active speculators increase the rate of information diffusion from the equity and commodity markets into FX strategies. Thus, speculators in the FX market should play an important role in cross-asset information diffusion, leading to the hypothesis that the predictive ability of the equity and commodity market returns on FX strategy returns is weaker when FX speculators are active.

We find that broad equity and commodity market returns predict the returns of the majority of the FX strategies. In line with our hypothesis, the results show that the predictability of FX strategy returns by the equity market evaporates when speculators are active. These findings are novel and highlight the importance of speculators in the FX market. In addition, we test for the gradual information diffusion hypothesis as the explanation of the predictability of FX strategy returns in the same fashion as Driesprong et al. (2008) and Lu and Jacobsen (2016), and we find evidence confirming that speculators enhances information diffusion across asset markets. Our additional analysis using hedger activity shows the opposite pattern, compared to speculator activity, that cross-asset predictability is only present when hedgers are active.

The commodity market, however, differs from the equity market in that the predictive effect of the commodity market on the FX market does not disappear when speculators are active. Thus, it is possible that speculators in the FX market do not monitor or exploit information from the commodity market. Alternatively, the predictive ability of the commodity market on the FX market could be driven by the time-varying risk premium of carry trade returns, consistent with Bakshi and Panayotov (2013) and Ready et al. (2017).

Overall, our analysis finds that speculators are important catalysts in the information diffusion process. This result is key to understanding information dissemination across seemingly distinct but economically linked asset markets. Further research into understanding the nature of this relationship also has merits. Moreover, models that examine market price behaviour should consider the role of speculators.

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